DAKDL Baby-sitting Service

Data Warehouse Design (Star Schema)

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1. The 4-Step Dimensional Design (Task 2)

Step 1 — Business Process. Babysitting job fulfilment: one babysitting job carried out for a customer by an employee at a Club. This single process yields the CFO's revenue and working-hour metrics, so one fact table suffices.

Step 2 — Grain. One row in the fact table = one babysitting job (one `Job` record from the OLTP source): one engagement, one date, one employee, one customer, one Club, one job type. This atomic grain rolls up to week / month / quarter / year.

Step 3 — Dimensions (Who / Where / When / What):

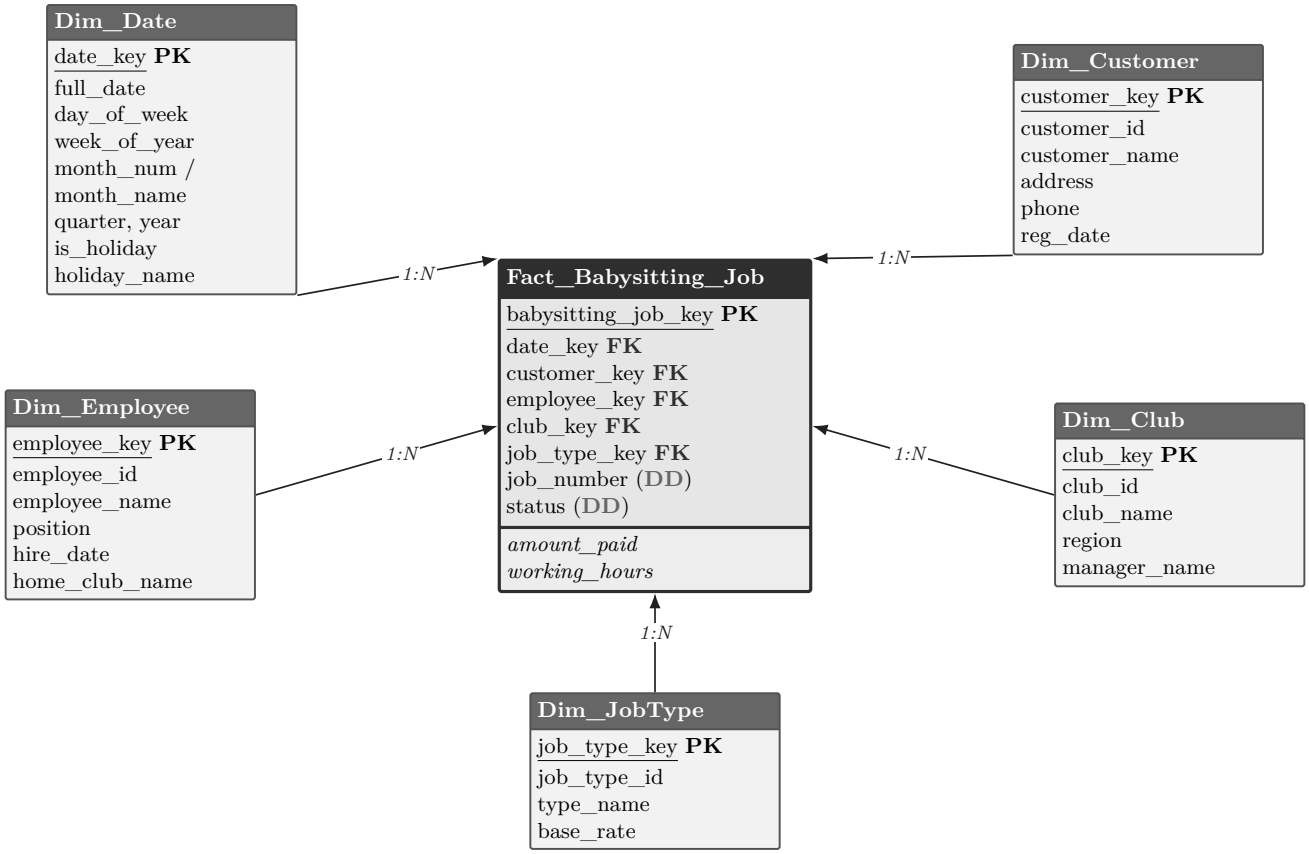
Dimension	Role	Provides
Dim_Customer	Who (pays)	Customer name, address, phone, registration date.
Dim_Employee	Who (works)	Employee name, position, hire date, home Club.
Dim_Club	Where	Branch name, region, manager.
Dim_Date	When	Day / week / month / quarter / year + holiday flag.
Dim_JobType	What	Type of care and its base hourly rate.

`job_number` and `status` are kept in the fact as **degenerate dimensions** (DD).

Step 4 — Facts (measures):

Measure	Additivity	Meaning
amount_paid	Additive	Revenue for the job (THB).
working_hours	Additive	Hours worked = `TimeEnd` − `TimeBegin`.

2. Star Schema Diagram (Task 3)



One central fact, five dimensions (a star). Each dimension links to the fact 1:N via its surrogate key (**PK** in the dimension, **FK** in the fact); `job_number` and `status` are degenerate dimensions (DD); measures sit in the fact's lower partition.